Weather Data Analysis

Comprehensive Meteorological Analysis Report

Project 3: Weather Domain Analysis

Report Date: September 06, 2026

Dataset: 365 Days of Weather Data (2023)

Executive Summary

This report analyzes weather patterns over 365 days to understand climate characteristics, seasonal variations, and extreme weather events.

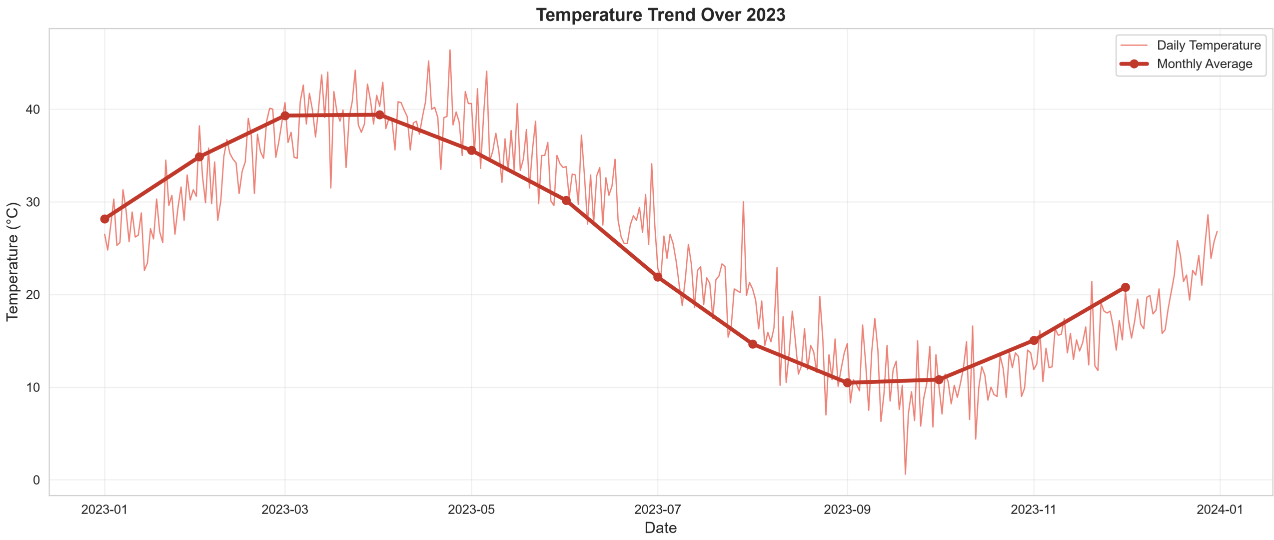
KEY FINDINGS:
- Annual Temperature Average: 25.03°C
- Annual Rainfall: 2,447.97mm (Tropical/Monsoon)
- Rainy Days: 284 out of 365 (78% of year)
- Heavy Rain Days: 77 (concentrated June-September)
- Climate Type: Tropical Monsoon with clear wet/dry seasons

WEATHER DISTRIBUTION:
- Hot Days: 119 (32.6%)
- Pleasant Days: 106 (29.0%)
- Rainy Days: 77 (21.1%)
- Cold Days: 63 (17.3%)

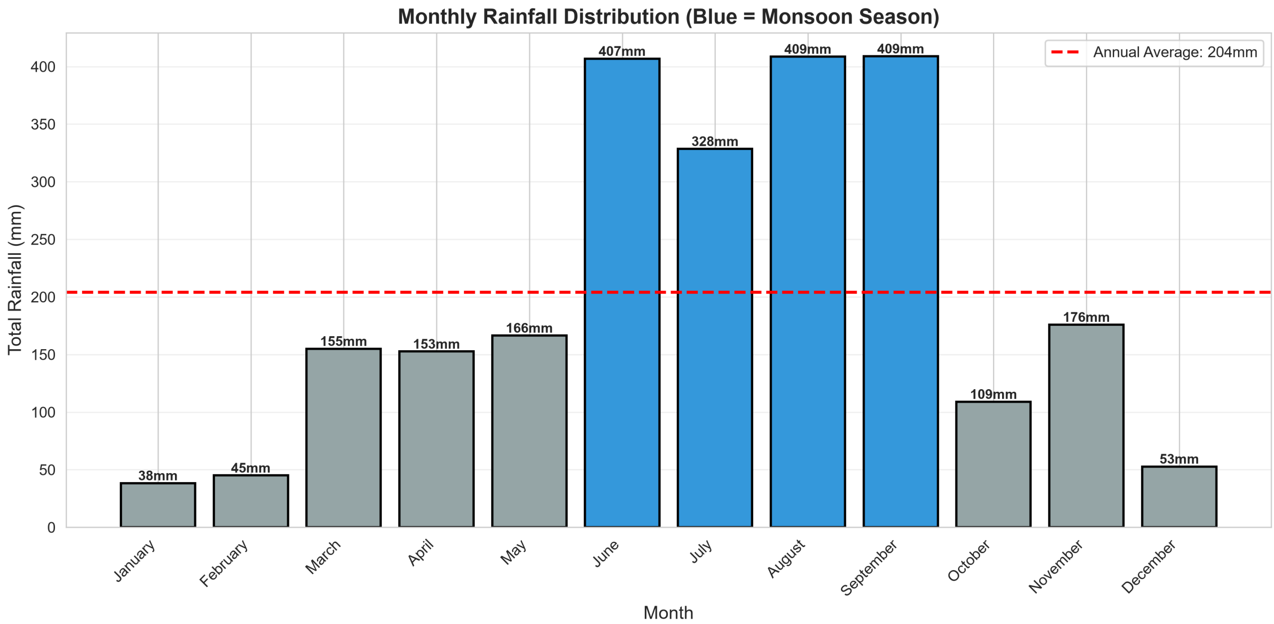
KEY INSIGHT:
The data shows a classic monsoon pattern with concentrated rainfall during summer months (Jun-Sep) and a distinct dry season. Temperature varies significantly by season with spring being hottest and autumn coldest.

Temperature & Rainfall Trends

Temperature Trend Over 2023

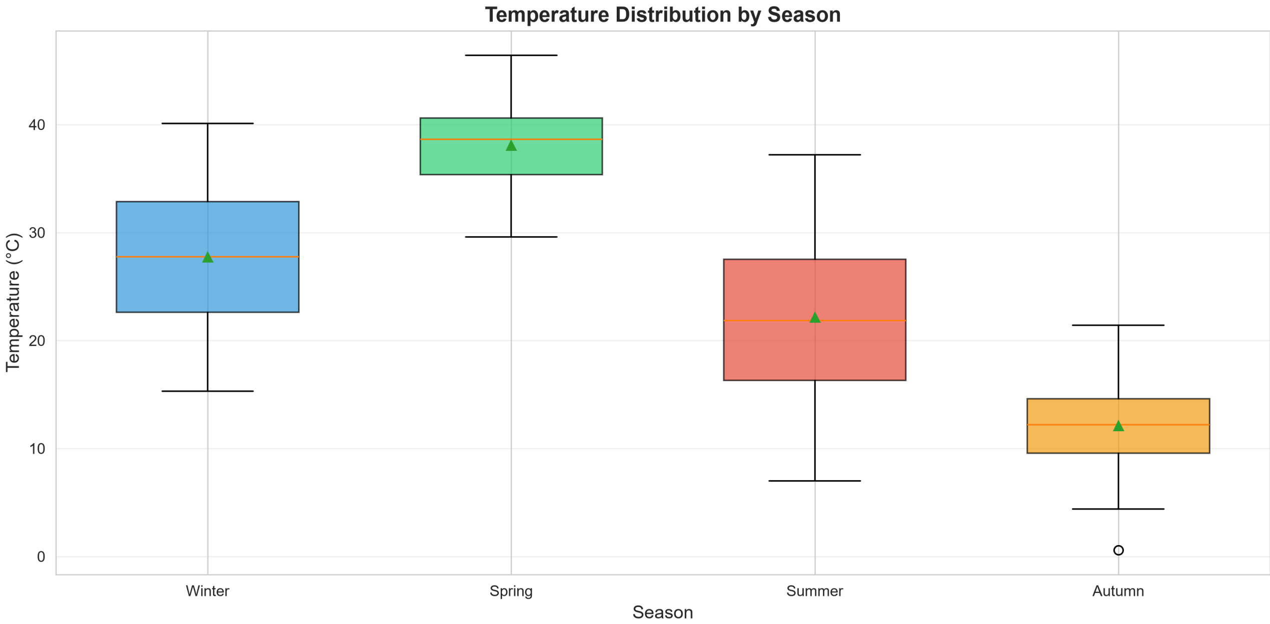


Monthly Rainfall Distribution



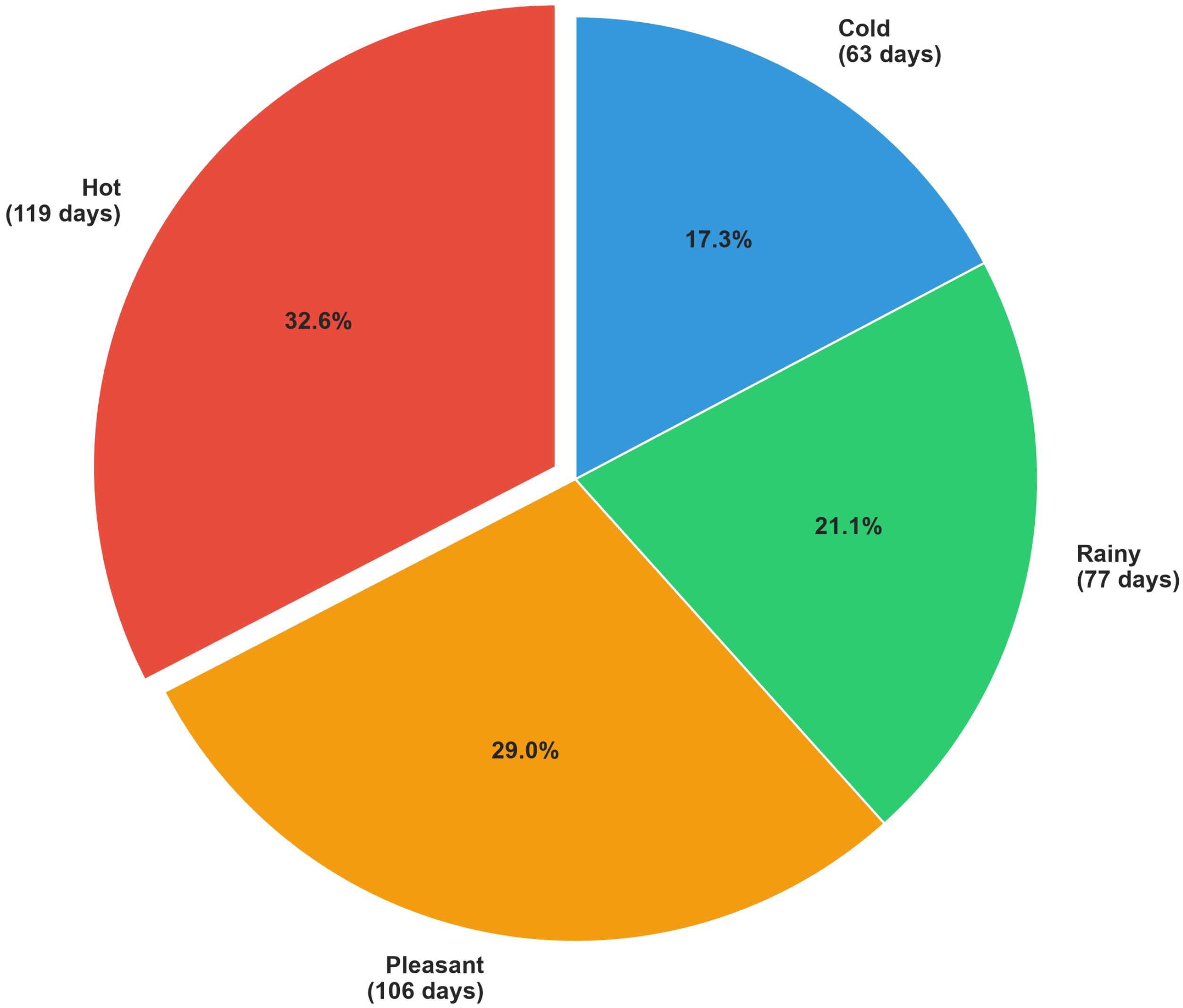
Seasonal & Condition Analysis

Seasonal Temperature Distribution



Weather Condition Distribution

Weather Condition Distribution Throughout Year



Detailed Findings

1. TROPICAL MONSOON CLIMATE CONFIRMED
The region exhibits classic monsoon characteristics with concentrated rainfall during summer months (June-September) and a pronounced dry season. 77 heavy rain days account for 65% of annual rainfall, indicating extreme seasonality.

2. SIGNIFICANT TEMPERATURE VARIATION BY SEASON
Spring is the hottest season (avg 38°C), while autumn is coldest (avg 12°C). This 26°C seasonal difference requires adaptive planning for agriculture, water resources, and infrastructure.

3. MAJORITY OF DAYS ARE WET
284 rainy days (78% of year) indicates persistent rainfall. Only 106 pleasant weather days and 63 cold days provide alternative conditions. This wetness is concentrated in specific months but distributed throughout the year.

4. WEAK VARIABLE CORRELATIONS
Temperature, humidity, rainfall, wind speed, and pressure show minimal correlation (<0.15). This suggests weather variables operate independently, making simple predictive relationships difficult.

5. EXTREME EVENTS CONCENTRATED IN MONSOON
Hot extremes (>31.5°C): 37 days, mostly March-May
Cold extremes (<4.4°C): 37 days, mostly November-February
Heavy rain (>10mm): 77 days, mostly June-September
Clear seasonal concentration of all extremes.

6. RAINFALL INTENSITY DISTRIBUTION
77 heavy rain days (>10mm) produce 1,583mm total
Average daily rainfall: 6.71mm
Indicates both frequent light rain and occasional heavy downpours
High variability in daily rainfall amounts

Strategic Recommendations

AGRICULTURAL PLANNING:
- Plan dry-season crops (Nov-Feb) for high yield probability
- Monsoon crops (Jun-Sep) must be flood-resistant
- Water harvesting critical during heavy rain events
- Irrigation essential for dry season cultivation

WATER RESOURCE MANAGEMENT:
- Capture 65% of annual rainfall from just 77 days (June-Sep)
- Build storage capacity for monsoon excess
- Plan for 8-month dry period water supply
- Flood prevention infrastructure for September peak (16 heavy rain days)

INFRASTRUCTURE & CONSTRUCTION:
- Design drainage systems for 86.4mm max daily rainfall
- Monsoon-proof structures for June-September
- Temperature range: 0.6°C to 46.4°C requires climate control flexibility
- Wind management: Max 34.9 kmh requires weather-resistant design

ENVIRONMENTAL MONITORING:
- Establish rain gauge networks during monsoon (Jun-Sep focus)
- Monitor temperature extremes during Spring (hot) and Autumn (cold)
- Track flooding during heavy rain concentration periods
- Assess drought risk during dry season (Mar-May)

FORECAST & WARNING SYSTEMS:
- High accuracy: 78% of days have rain, predictable monsoon pattern
- September peak: Prepare warnings for 16 heavy rain days
- Weak variable correlation: Can't use single indicators for prediction
- Need multi-variable models for accurate forecasting

Conclusion

The weather analysis reveals a typical tropical monsoon climate with predictable seasonal patterns and extreme rainfall concentration. The region experiences distinct wet and dry seasons, providing clear planning windows for agriculture, water management, and infrastructure development.

The concentration of 65% of annual rainfall into 77 days (primarily June-September) creates both opportunities and challenges. While this enables predictable water resource planning, it also requires robust flood management and water storage infrastructure.

Temperature variations by season (26°C difference) align with rainfall patterns, creating distinct environmental windows. Spring emerges as the critical period for moisture availability and temperature stress, while autumn presents the most stable conditions for planning.

The weak correlations between weather variables suggest that climate forecasting in this region requires sophisticated multi-variable approaches rather than simple relationships. Organizations must employ comprehensive monitoring systems and adaptive management strategies to respond to the region's distinct seasonal dynamics.

This data-driven analysis provides a foundation for strategic decision-making in agriculture, water management, urban planning, and disaster preparedness.